



<- This means this lab is on your local computer

Lab 6: CPU, Pipelining

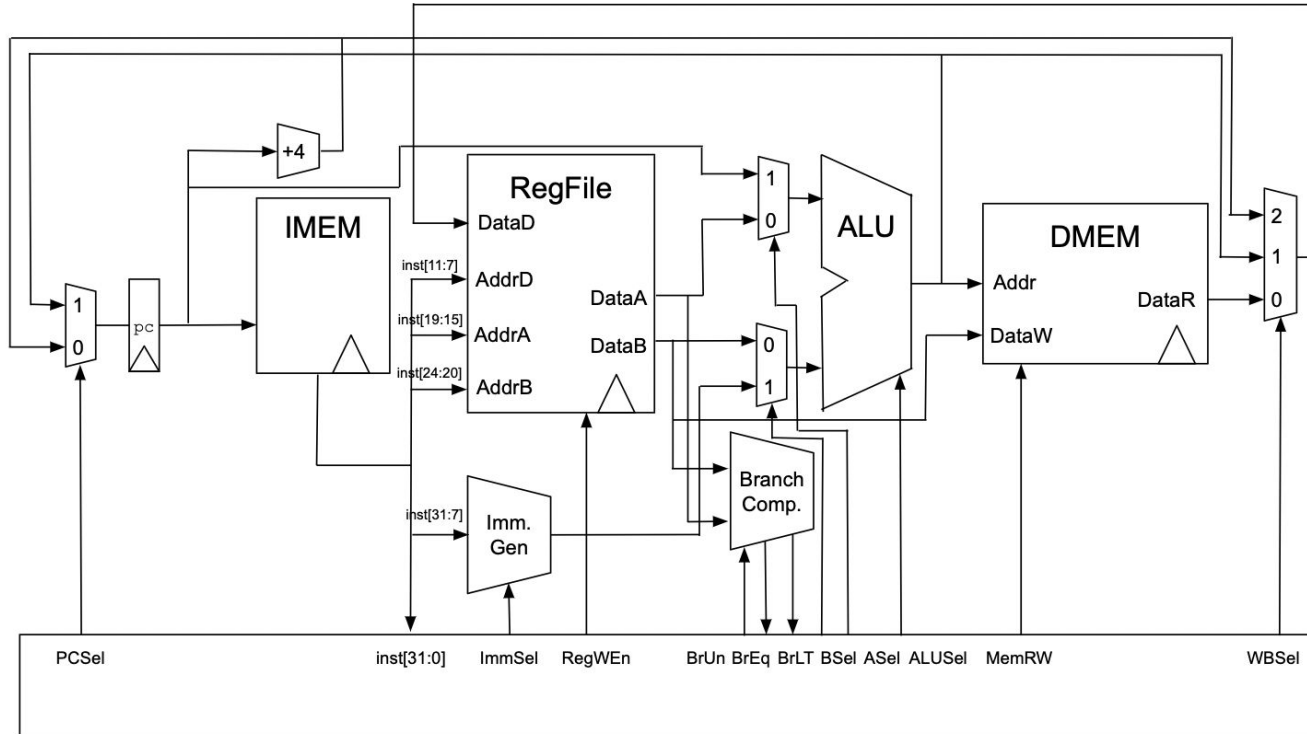
CS61C SU26

Immediates

	31	25 24	20 19	15 14	12 11	7 6	0
R	funct7	rs2	rs1	funct3	rd	opcode	
I	imm[11:0]			rs1	funct3	rd	opcode
I*	funct7	imm[4:0]	rs1	funct3	rd	opcode	
S	imm[11:5]	rs2	rs1	funct3	imm[4:0]	opcode	
B	imm[12 10:5]	rs2	rs1	funct3	imm[4:1 11]	opcode	
U	imm[31:12]					rd	opcode
J	imm[20 10:1 11 19:12]					rd	opcode

Immediates are sign-extended to 32 bits, except in I* type instructions and **sltiu**.

Control Signals



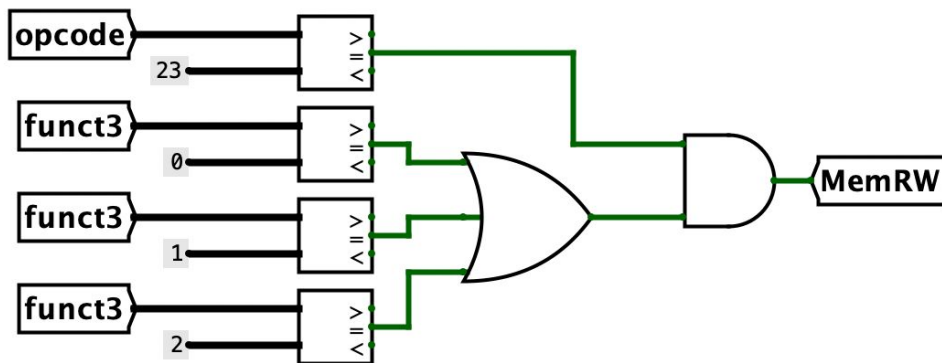
MemRW Example

MemRW Value	Occurs when
0	Reading from memory
1	Writing to memory
Don't Care	Can never be don't care!

Instruction	Name	Description	Type	Opcode	Funct3
sb rs2 imm(rs1)	Store Byte	Stores least-significant byte of rs2 at the address rs1 + imm in memory	S	010 0011	000
sh rs2 imm(rs1)	Store Half-word	Stores the 2 least-significant bytes of rs2 starting at the address rs1 + imm in memory	S	010 0011	001
sw rs2 imm(rs1)	Store Word	Stores rs2 starting at the address rs1 + imm in memory	S	010 0011	010

MemRW Example

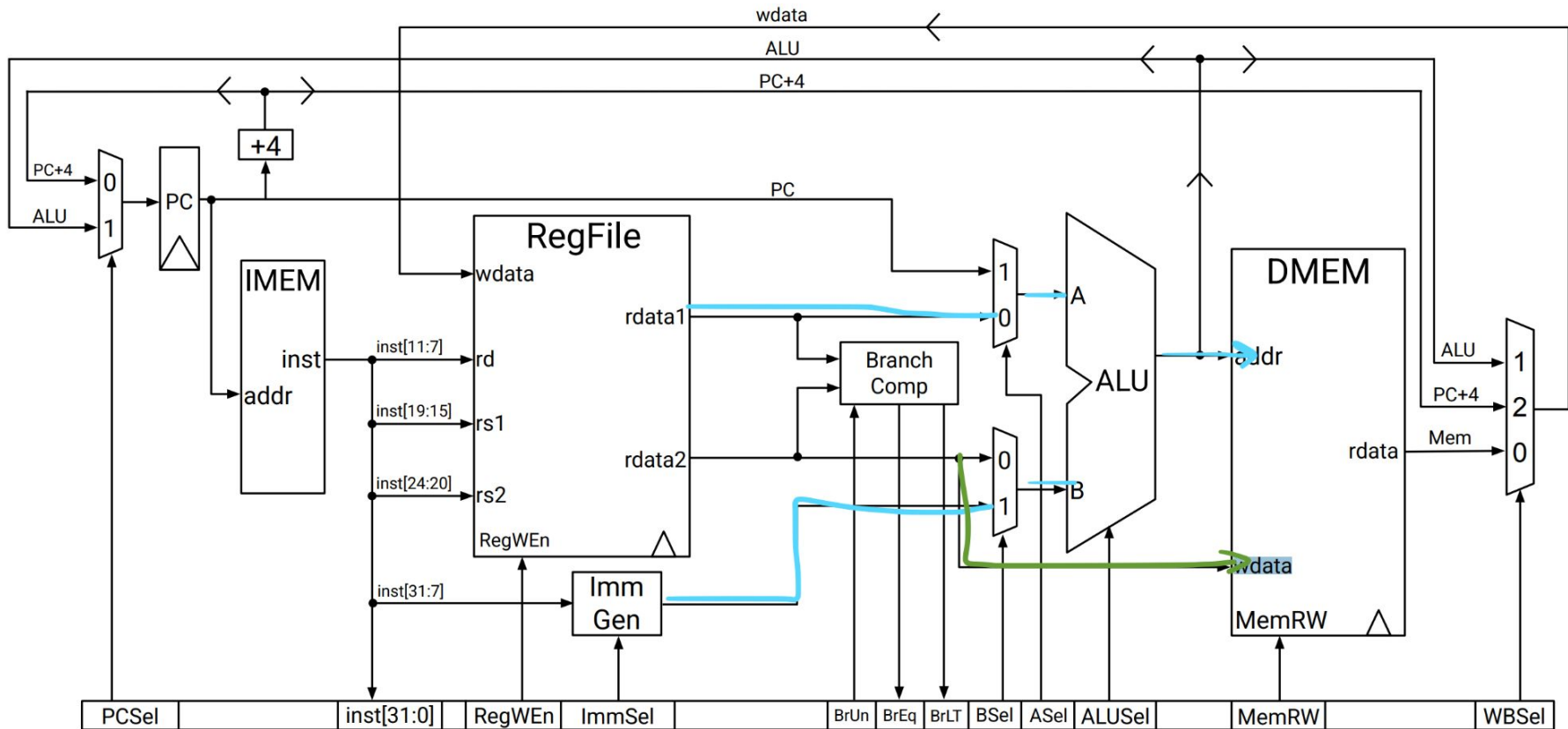
Instruction	Name	Description	Type	Opcode	Funct3
sb rs2 imm(rs1)	Store Byte	Stores least-significant byte of rs2 at the address rs1 + imm in memory	S	010 0011	000
sh rs2 imm(rs1)	Store Half-word	Stores the 2 least-significant bytes of rs2 starting at the address rs1 + imm in memory	S	010 0011	001
sw rs2 imm(rs1)	Store Word	Stores rs2 starting at the address rs1 + imm in memory	S	010 0011	010



BrUn

BrUn Value	Occurs when
0	Signed branch comparison
1	Unsigned branch comparison
Don't Care	beq, bne, non-branch instruction

Instruction	Name	Description	Type	Opcode	Funct3
beq rs1 rs2 label	Branch if Equal	if (rs1 == rs2) PC = PC + offset	B	110 0011	000
bge rs1 rs2 label	Branch if Greater or Equal (signed)	if (rs1 >= rs2) PC = PC + offset	B	110 0011	101
→ bgeu rs1 rs2 label	Branch if Greater or Equal (Unsigned)	PC = PC + offset	B	110 0011	111
blt rs1 rs2 label	Branch if Less Than (signed)	if (rs1 < rs2) PC = PC + offset	B	110 0011	100
→ bltu rs1 rs2 label	Branch if Less Than (Unsigned)	PC = PC + offset	B	110 0011	110
bne rs1 rs2 label	Branch if Not Equal	if (rs1 != rs2) PC = PC + offset	B	110 0011	001



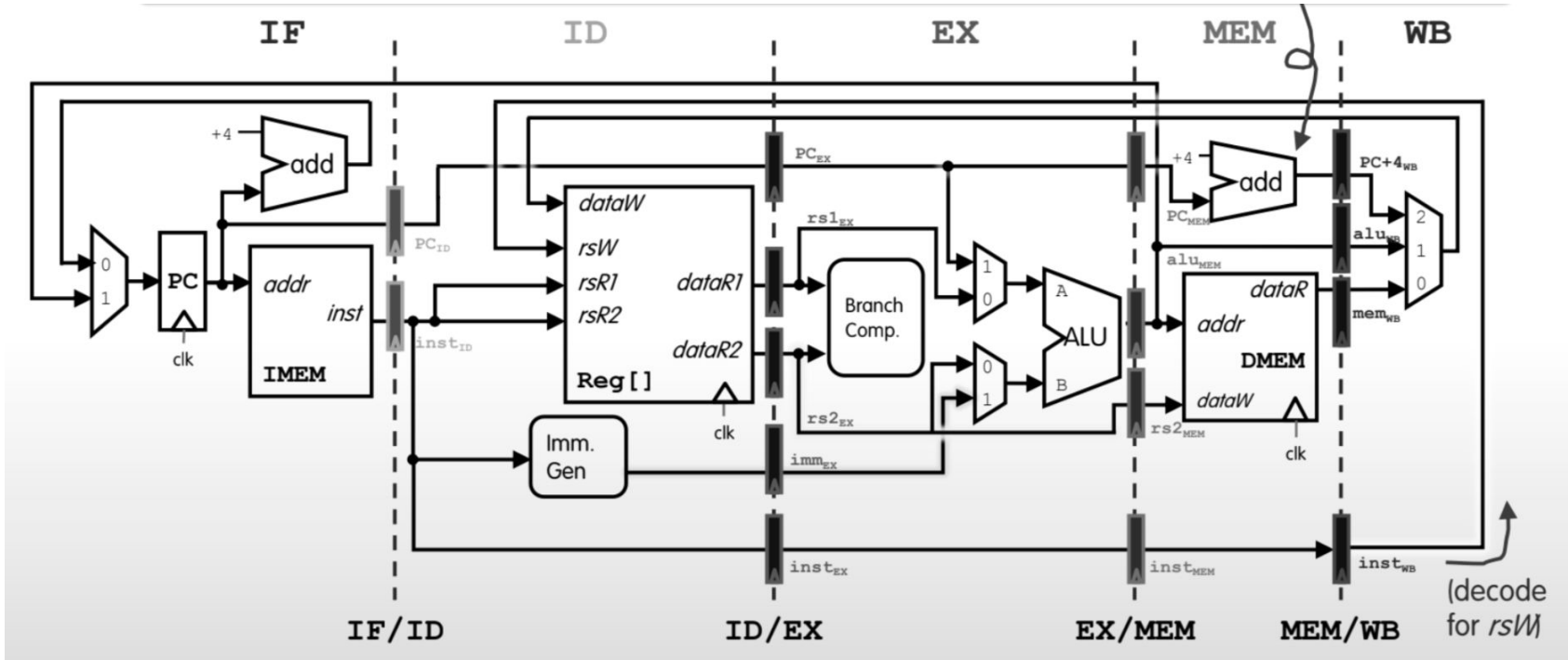
Critical Path

- The longest path in the circuit
- Limits the maximum clock speed
- Can be measured from:
 - register to register
 - register to output (since the output typically also goes to a register)
 - input to register (since the input typically also comes from a register)
- $\text{clock period} \geq \text{clk-to-q} + \text{max CL delay} + \text{setup time}$
 - Maximum cumulative delay of all combinational logic through a path (max CL delay)...
 - coming from the output of a register (clk-to-q)...
 - and ending at the input of another register (setup time)
 - Clock period must be *long enough* so that the longest delay (clk-to-q + max CL delay) is accounted for while satisfying the register timing requirement (setup time)
- Clock Frequency (Hz = cycles/s): $1/(\text{clock_period})$ *Units!!
- When calculating delays, you don't pass through a register!

Throughput vs Latency

- Latency is how long it takes to complete a single task
- Throughput is how many tasks can be completed per unit time

Pipelining



Does pipelining improve latency or throughput?

Only throughput because every component of the pipeline still takes the same amount of time, we just don't have to wait for one instruction to finish before starting the next one